

Graphing Basic Polar Curves from Equations

Precalculus

Evaluating r at a given angle, finding the angles where r is zero, locating the maximum r that fixes a curve's orientation, and identifying and sketching basic polar curves.

- Work in degrees. Where an interval is asked for, use $0^\circ \leq \theta < 360^\circ$ and list every solution in that window.
- Give exact values, not decimals.
- A negative r is legal: plot $|r|$ in the direction opposite θ .
- Show the equation you solved, not just the angles.

1. For the circle $r = 6 \cos \theta$, find r when $\theta = 60^\circ$.

Answer: _____

2. For the circle $r = 4 \sin \theta$, find r when $\theta = 30^\circ$.

Answer: _____

3. For the cardioid $r = 1 + \cos \theta$, find r when $\theta = 90^\circ$. This value is the curve's distance from the pole straight up the 90° ray.

Answer: _____

4. The rose $r = 4 \sin(2\theta)$ returns to the pole between petals. Find every θ in $0^\circ \leq \theta < 360^\circ$ for which $r = 0$. These angles are the lines the petals are tucked between.

Answer: _____

5. For the limaçon $r = 2 + 4 \cos \theta$, find r when $\theta = 180^\circ$. Read your answer carefully: the sign matters.

Answer: _____

- 6.** For the cardioid $r = 1 + \cos \theta$, find every θ in $0^\circ \leq \theta < 360^\circ$ with $r = 0$. This single angle is where the cardioid's cusp sits.

Answer: _____

- 7.** The rose $r = 2 \sin(3\theta)$ reaches its greatest distance from the pole when $r = 2$. Solve $2 \sin(3\theta) = 2$ for every θ in $0^\circ \leq \theta < 360^\circ$. These angles point along the petal tips.

Answer: _____

8. For the limaçon $r = 3 - 2 \cos \theta$, find r when $\theta = 120^\circ$.

Answer: _____

9. The limaçon $r = 2 + 4 \cos \theta$ passes through the pole twice, and those two angles bound its inner loop.

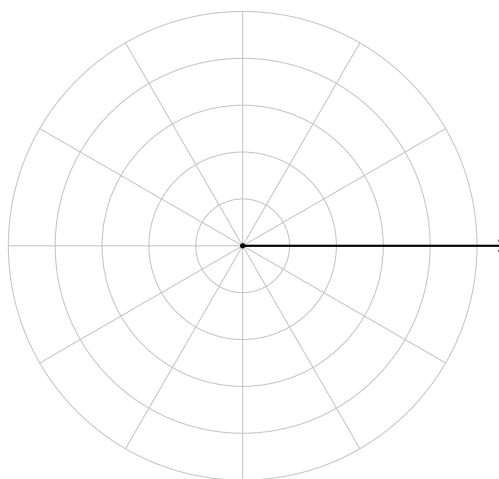
Find every θ in $0^\circ \leq \theta < 360^\circ$ for which $r = 0$.

Answer: _____

10. For the circle $r = 4 \cos \theta$, the maximum value of r is 4. Solve $4 \cos \theta = 4$ on $0^\circ \leq \theta < 360^\circ$. The single angle you find is the direction in which the circle is offset from the pole.

Answer: _____

11. Sketch. Graph the cardioid $r = 1 + \cos \theta$ on the polar grid below. Plot the four points at $\theta = 0^\circ, 90^\circ, 180^\circ$, and 270° first, then join them smoothly. Label the cusp. Each ring is 1 unit; the outermost ring is $r = 5$. The heavy ray is the polar axis, $\theta = 0^\circ$.



12. Identify, sketch, and justify. Consider $r = 2 + 4 \cos \theta$.

Name the family this curve belongs to, sketch it on the grid below, and explain — using the constant term and the coefficient of $\cos \theta$ — why this curve has an inner loop while $r = 4 + 2 \cos \theta$ would not. Use the pole-crossing angles you found earlier as evidence.

Each ring is 1 unit; the outermost ring is $r = 6$. The heavy ray is the polar axis, $\theta = 0^\circ$.

